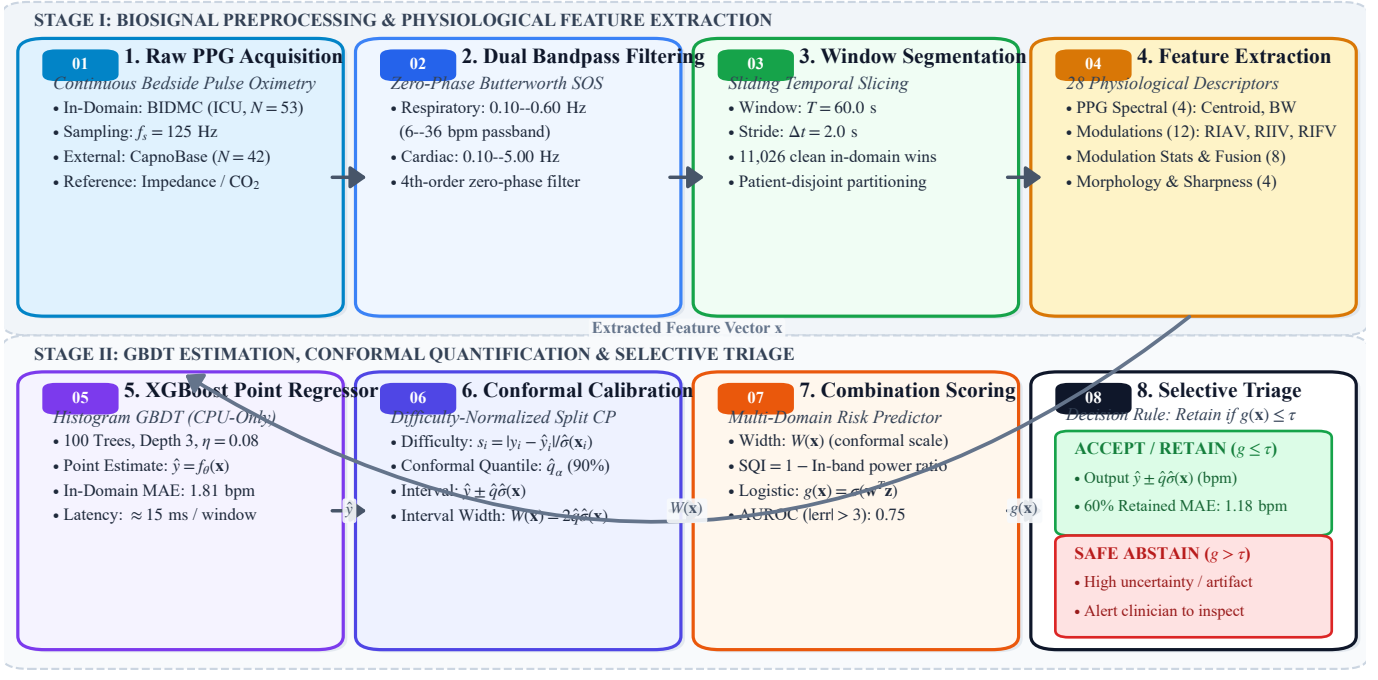

STAGE I: BIOSIGNAL PREPROCESSING & PHYSIOLOGICAL FEATURE EXTRACTION
01
1. Raw PPG Acquisition
Continuous Bedside Pulse Oximetry
• In-Domain: BIDMC (ICU, N = 53)
• Sampling: f_s = 125 Hz
• External: CapnoBase (N = 42)
• Reference: Impedance / CO2
02
2. Dual Bandpass Filtering
Zero-Phase Butterworth SOS
• Respiratory: 0.10--0.60 Hz
(6--36 bpm passband)
• Cardiac: 0.10--5.00 Hz
• 4th-order zero-phase filter
03
3. Window Segmentation
Sliding Temporal Slicing
• Window: T = 60.0 s
• Stride: Δt = 2.0 s
• 11,026 clean in-domain wins
• Patient-disjoint partitioning
04
4. Feature Extraction
28 Physiological Descriptors
• PPG Spectral (4): Centroid, BW
• Modulations (12): RIAV, RIIV, RIFV
• Modulation Stats & Fusion (8)
• Morphology & Sharpness (4)
Extracted Feature Vector x
STAGE II: GBDT ESTIMATION, CONFORMAL QUANTIFICATION & SELECTIVE TRIAGE
05
5. XGBoost Point Regressor
Histogram GBDT (CPU-Only)
• 100 Trees, Depth 3, η = 0.08
• Point Estimate: ŷ = f_θ(x)
• In-Domain MAE: 1.81 bpm
• Latency: ≈ 15 ms / window
ŷ
06
6. Conformal Calibration
Difficulty-Normalized Split CP
• Difficulty: s_i = |y_i − ŷ_i|/σ̂(x_i)
• Conformal Quantile: q̂_α (90%)
• Interval: ŷ ± q̂σ̂(x)
• Interval Width: W(x) = 2q̂σ̂(x)
W(x)
07
7. Combination Scoring
Multi-Domain Risk Predictor
• Width: W(x) (conformal scale)
• SQI = 1 − In-band power ratio
• Logistic: g(x) = σ(w⃗ᵀ z⃗)
• AUROC (|err| > 3): 0.75
g(x)
08
8. Selective Triage
Decision Rule: Retain if g(x) ≤ τ
ACCEPT / RETAIN (g ≤ τ)
• Output ŷ ± q̂σ̂(x) (bpm)
• 60% Retained MAE: 1.18 bpm
SAFE ABSTAIN (g > τ)
• High uncertainty / artifact
• Alert clinician to inspect